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Roll No

FT-222-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Strength of Materials

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define Thermal stresses and Thermal strength.
b) A steel tube of 4.6cm external diameter and 3mm thickness encloses centrally a solid copper bar of 3cm diameter. The bar and the tube are rigidly connected together at the ends at a temperature of 30°C. Find the stress in each metal when heated to 160°C. Also find the increase in length if the original length of the assembly is 30cm.
Take $\alpha_s = 1.08 \times 10^{-5} / ^\circ\text{C}$, $\alpha_c = 1.7 \times 10^{-5} / ^\circ\text{C}$, $E_s = 210 \text{ GPa}$, $E_c = 110 \text{ GPa}$.
2. A thick cylindrical pressure vessel of inner radius 150mm is subjected to an internal pressure of 80Mpa. Calculate the wall thickness based upon the
 - i) Maximum principal stress theory
 - ii) Maximum strain energy theoryTake Poisson's ratio = 0.30 and yield strength = 300 Mpa.
3. a) Discuss different theories of Failure in brief. Graphically represent these theories.
b) A solid circular shaft is subjected to a torque of 15 kNm. Determine the diameter of shaft using maximum shear stress theory.

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4.
 - a) Explain in detail maximum principal strain theory and maximum strain energy theory.
 - b) What do you mean by critical load of column? Write formula for Euler's critical load for each condition.
5.
 - a) Derive an expression for the slope and deflection of a beam subjected to uniform bending moment.
 - b) Write short notes on Moment Area Method.
6.
 - a) Prove that the relation $M = EI \frac{d^2 y}{dx^2}$
Where M = Bending Moment E = Young's modulus
I = Moment of Inertia.
 - b) Derive the flexure formula $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$, where symbol represents the usual meaning.
7. At a section of a mild steel shaft, the maximum torque is 8437.5 Nm and maximum bending moment is 5062.5 Nm. The diameter of shaft is 90 mm and the stresses at the elastic limit in simple tension for the material of the shaft is 220 N/mm². Determine whether the failure of the material will occur or not according to maximum shear stress theory, if not then find factor of safety.
8. Write short notes on:
 - a) Torsion and Torsional rigidity
 - b) Equivalent length of column
 - c) Section modulus
 - d) Neutral axis

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